# HBV-EC Input Parameter Guide

Guichon Creek at Outlet of Mamit Lake (WSC 08LG041). Raven v4.1 HBV-EC emulation. Project SWC-1-43-01, m.ah a temEEwuh Solar.

This guide lists every value that still has to be collected or decided before the model can run and be calibrated. For each one it gives the file, the block or command, and the field where it goes. All paths are relative to `Raven/HBVEC_Model/`. In the files, lines marked `>>> TODO` hold placeholder values that must be replaced. Rows shaded green are already done.

## 1. Status by file

| File | Holds | Data source | Status | Notes |
|---|---|---|---|---|
| `data/obs_flow_daily.rvt` | Observed daily flow at subbasin 1 | WSC HYDAT | Done | 08LG041, 1982-01-01 to 2024-12-31, 439 missing days (-1.2345). Rebuild with `python3 data/get_obs_08LG041.py`. Flags in `data/08LG041_daily_HYDAT.csv`. |
| `hbvec.rvh` | Subbasin and HRUs | GIS: DEM, land cover, basin outline | To do | All six HRU rows are placeholders. HRU areas must sum to 870.1 km$^2$. |
| `data/met_daily.rvt` | Daily Tmax, Tmin, precipitation | Climate station or gridded data | To do | Empty (N = 0). Must be gap-free over the whole run period. |
| `hbvec.rvt` | Met gauge location, monthly normals, gauge corrections | Same source as the met data | To do | Location and normals are placeholders. |
| `hbvec.rvi` | Run period and evaluation windows | You choose | Check | Currently 2000-10-01 to 2020-09-30. Confirm once the met record is known. |
| `hbvec.rvc` | Initial storages | Rough estimates | Check | Set the initial Q from the observed flow, and give one SOIL[2] value per HRU. |
| `hbvec.rvp` | Model parameters | Tutorial starting values, then calibration | Calibrate | See Section 6. A few land-cover values need a decision for the post-project scenario (Section 7). |

## 2. Watershed and HRUs: `hbvec.rvh`

The model is lumped: one subbasin, split into elevation bands, with each band split into OPEN and FOREST HRUs. Add or remove HRU rows to suit the basin. Derive everything from the basin polygon delineated to the 08LG041 gauge.

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Basin name | `:SubBasins` | NAME (row 1) | text | Use a short name with no spaces, e.g. `Guichon_08LG041`. Leave GAUGED = 1 and DOWNSTREAM_ID = -1. |
| Gauge drainage area (check value) | n/a (QA check) | Sum of HRU AREA | km$^2$ | **870.1 km$^2$** (WSC gross drainage area). The delineated polygon should match within a few percent. |
| Gauge location | n/a (for delineation) | n/a | deg | 08LG041 is at 50.3611 N, 120.8099 W. Delineate the basin to this point. |
| Elevation bands | `:HRUs` | one group of rows per band | m | From the DEM. Use 200-500 m bands, or 3-10 bands in total. |
| HRU area | `:HRUs` | AREA | km$^2$ | Band area x open or forest fraction. The total must equal the drainage area. |
| HRU mean elevation | `:HRUs` | ELEVATION | m | Mean DEM elevation inside the HRU. |
| HRU centroid | `:HRUs` | LATITUDE, LONGITUDE | deg | HRU or basin centroid, roughly 50.4 N, -120.8 E. Longitude is negative (west). |

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Land cover class | `:HRUs` | LAND_USE_CLASS, VEG_CLASS | class | OPEN or FOREST, from land cover (e.g. BC VRI, NRCan land cover). Decide whether burnt stands count as FOREST or OPEN (see Section 7). |
| Mean slope | `:HRUs` | SLOPE | deg | From the DEM. Used by the HBV aspect melt correction. |
| Mean aspect | `:HRUs` | ASPECT | deg | Clockwise from north (0 = N, 180 = S). Take a circular mean of the DEM aspect. |
| Lakes | `:HRUs` | SOIL_PROFILE = LAKE | $km^2$ | Mamit Lake and other open water. Treat them as LAKE HRUs; lake storage damps the peaks at the gauge. |
| Glaciers | `:HRUs` | LAND_USE_CLASS = GLACIER | n/a | Probably none in this basin. If so, keep row 7 commented out. |
| HRU group range | `:HRUGroup AllHRUs` | ID range | IDs | Update `1-6` if the number of HRUs changes. |
| Routing (MAXBAS) | `:SubBasinProperties` | TIME_CONC, TIME_TO_PEAK | d | Calibrated. Start with 2.0 and 1.0, and keep TIME_TO_PEAK at TIME_CONC / 2. |

## 3. Daily meteorological forcing: `data/met_daily.rvt`

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Daily maximum temperature | `:MultiData` | TEMP_MAX column | deg C | From an ECCC climate station near Logan Lake / Highland Valley, or gridded data (Daymet, ERA5-Land). |
| Daily minimum temperature | `:MultiData` | TEMP_MIN column | deg C | Same source. |
| Daily total precipitation | `:MultiData` | PRECIP column | mm/d | Rain plus snow water equivalent. Raven splits it with RAINSNOW_HBV. |
| Record header | `:MultiData` | start date, time step (1.0), N rows | n/a | N must equal the number of data rows. The record must cover :StartDate to :EndDate in hbvec.rvi. |
| Gap filling | n/a | n/a | n/a | Forcing **cannot have gaps**. Fill them from a nearby station or gridded data before you write the file. |

## 4. Met gauge and monthly normals: `hbvec.rvt`

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Gauge latitude | `:Gauge MetStation` | `:Latitude` | deg | Station location, or the grid cell or basin centroid for gridded data. |
| Gauge longitude | `:Gauge MetStation` | `:Longitude` | deg | Negative in the western hemisphere. |
| Gauge elevation | `:Gauge MetStation` | `:Elevation` | m | **Important.** Temperature and precipitation are lapsed from this elevation to each HRU. |
| Monthly mean temperature | `:MonthlyAveTemperature` | 12 values, Jan-Dec | deg C | Calculate from the Step 3 data, or take from ECCC climate normals. |
| Monthly mean PET | `:MonthlyAveEvaporation` | 12 values, Jan-Dec | mm/d | **mm/day, not mm/month.** Use published PET or lake-evaporation normals, or calculate from temperature (Hargreaves, Oudin). |
| Rain undercatch (RFCF) | `:RainCorrection` | multiplier | - | Calibrated. Start at 1.0. |
| Snow undercatch (SFCF) | `:SnowCorrection` | multiplier | - | Calibrated. Start at 1.0. |
| Extra met stations (optional) | additional `:Gauge` blocks | n/a | n/a | With INTERP_NEAREST_NEIGHBOR, each HRU uses the gauge closest to it. |

## 5. Run period and initial conditions: `hbvec.rvi`, `hbvec.rvc`

| Value to decide | File: block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Simulation start / end | rvi: `:StartDate`, `:EndDate` | date | date | Must fall within both the met record and the flow record (flow: 1982-01-01 to 2024-12-31). Start on Oct 1. |
| Calibration window | rvi: `:EvaluationPeriod CALIBRATION` | start, end | date | Starts after a warm-up of at least one year (two is better). |
| Validation window | rvi: `:EvaluationPeriod VALIDATION` | start, end | date | An independent period. Note the flow gaps in Dec 2001 - Mar 2002 and May 2010. |
| Initial outlet flow | rvc: `:BasinInitialConditions` | Q | $m^3/s$ | The observed flow on :StartDate, from `data/08LG041_daily_HYDAT.csv`. |
| Initial lower-zone storage | rvc: `:InitialConditions SOIL[2]` | one value per HRU | mm | The count must match the number of HRUs. Use 0 for LAKE or GLACIER HRUs. The warm-up year makes the values unimportant. |

## 6. Calibration parameters: `hbvec.rvp` (plus rvt and rvh)

These are the 21 classic HBV-EC parameters, with the HBV_PARA index used in the Salmon_HBV tutorial. Start values are the ones currently in the files. The ranges are *indicative*, taken from common HBV practice; adjust them to your judgement or your OSTRICH setup. **Bold** marks the parameters that usually matter most.

| PARA | HBV name | Raven parameter | Where (block / row) | Start | Range | Units |
|---|---|---|---|---|---|---|
| 01 | **TT** | `RAINSNOW_TEMP` | `:GlobalParameter` | 0.0 | -3 to 3 | deg C |
| 02 | **CFMAX** | `MELT_FACTOR` | `:LandUseParameterList` [DEFAULT] | 4.0 | 1.5 to 8 | mm/d/K |
| 03 | CFR | `REFREEZE_FACTOR` | `:LandUseParameterList` [DEFAULT] | 2.0 | 0 to 4 | mm/d/K |
| 04 | CWH | `SNOW_SWI` | `:GlobalParameter` | 0.05 | 0.01 to 0.25 | - |
| 05 | **FC** (via porosity) | `POROSITY` | `:SoilParameterList` [DEFAULT] | 0.2 | 0.05 to 0.5 | - |
| 06 | **LP** | `FIELD_CAPACITY` | `:SoilParameterList` [DEFAULT] | 0.7 | 0.3 to 1.0 | - |
| 07 | **BETA** | `HBV_BETA` | `:SoilParameterList` [DEFAULT] | 2.0 | 1 to 6 | - |
| 08 | **PERC** | `MAX_PERC_RATE` | `:SoilParameterList` FAST_RES | 5.0 | 0 to 10 | mm/d |
| 09 | **KF** | `BASEFLOW_COEFF` | `:SoilParameterList` FAST_RES | 0.2 | 0.01 to 0.5 | 1/d |
| 10 | **KS** | `BASEFLOW_COEFF` | `:SoilParameterList` SLOW_RES | 0.05 | 0.001 to 0.1 | 1/d |
| 11 | **MAXBAS** | `TIME_CONC` | rvh `:SubBasinProperties` | 2.0 | 1 to 7 | d |
| 12 | PCALT | `PRECIP_LAPSE` | `:GlobalParameter` | 0.0 | calibrate if multi-band | mm/d/km |
| 13 | TCALT | `ADIABATIC_LAPSE` | `:GlobalParameter` | 6.0 | 4 to 9 | deg C/km |
| 14 | wilting fraction | `SAT_WILT` | `:SoilParameterList` [DEFAULT] | 0.0 | 0 to 0.2 | - |
| 15 | ALPHA (1 + a) | `BASEFLOW_N` | `:SoilParameterList` FAST_RES | 1.5 | 1 to 3 | - |
| 16 | CFLUX | `MAX_CAP_RISE_RATE` | `:SoilParameterList` [DEFAULT] | 1.0 | 0 to 2 | mm/d |
| 17 | **FC** (via depth) | TOPSOIL thickness | `:SoilProfiles` DEFAULT_P | 1.0 | 0.5 to 3 | m |
| 18 | MRF | `HBV_MELT_FOR_CORR` | `:LandUseParameterList` [DEFAULT] | 0.5 | 0.3 to 1.0 | - |
| 19 | KG | `GLAC_STORAGE_COEFF` | `:LandUseParameterList` | 0.1 | n/a (no glacier) | 1/d |
| 20 | **RFCF** | `:RainCorrection` | rvt `:Gauge` | 1.0 | 0.8 to 1.5 | - |
| 21 | **SFCF** | `:SnowCorrection` | rvt `:Gauge` | 1.0 | 0.8 to 1.8 | - |

FC [mm] = TOPSOIL thickness [m] x 1000 x POROSITY. For example, 1.0 m x 1000 x 0.20 = 200 mm. Usually you calibrate one of the two and fix the other. The constants MIN_MELT_FACTOR (2.2), HBV_MELT_ASP_CORR (0.48), RAINSNOW_DELTA (2.0) and the glacier constants normally stay fixed. To have Raven list every parameter it expects, uncomment `:CreateRVPTemplate` in hbvec.rvi once.

## 7. Land-cover values for the baseline and post-project scenarios

The assessment compares current land cover with post-clearing land cover. The 260 ha solar footprint changes from forest or burnt forest to cleared or open. About 80% of the footprint, roughly **2.1 km$^2$**, lies inside the 08LG041 catchment. Collect the values below so that both scenarios can be built.

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Footprint polygon, clipped to the catchment | n/a (GIS) | n/a | km$^2$ | Intersect the footprint with the 08LG041 basin. Confirm the in-basin area (about 2.08 km$^2$). |

| Value to collect | Block / command | Field | Units | Source and notes |
|---|---|---|---|---|
| Footprint area by elevation band and cover | `:HRUs` | AREA | $km^2$ | How much of the footprint is FOREST, burnt or OPEN now, in each band. This sets how area moves between HRUs. |
| Burnt-forest treatment | `:HRUs` | LAND_USE_CLASS / VEG_CLASS | class | Decide whether the burnt stands are FOREST, OPEN, or a separate BURNT class. A separate class needs rows in every rvp table. |
| Post-project cover | rvh `:HRUs` | AREA (new or changed rows) | $km^2$ | Move the footprint area from the FOREST HRUs to OPEN, or to a new CLEARED HRU per band. Keep the total at 870.1 $km^2$. |
| Cleared-class properties (if a new class) | rvp `:LandUseClasses`, `:VegetationClasses`, `:VegetationParameterList` | FOREST_COV, MAX_HT, MAX_LAI, TFRAIN, TFSNOW | mixed | FOREST_COV = 0 means no MRF forest melt reduction. Interception fractions are TFRAIN and TFSNOW. |
| Panel and soil effects (optional) | rvp `:SoilParameterList` | e.g. a separate soil profile | n/a | Only if you want to represent compaction or panel drip-line concentration; this goes beyond the standard HBV-EC structure. |

## 8. Run and check

Run `Raven.exe hbvec -o output/` from `Raven/HBVEC_Model`. Read `output/Raven_errors.txt` first, then `Hydrographs.csv` and `Diagnostics.csv` (NSE, KGE, PBIAS, RMSE over the evaluation periods). Use `ForcingFunctions.csv` to check that PET, temperature and precipitation look sensible. Because the 08LG041 winter record has many ice-affected values, review the winter fit with that in mind.